AMLESH SWARUP SAHOO

Electrical Engineering Student | *Embedded Systems* | *Analog and PCB Design*
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EDUCATION

BITS Pilani and Iowa State University | 2+2 dual-degree program 2024–2028
B.E., Electronics and Communication Engineering; B.S., Electrical Engineering
Relevant labs: Digital Design; Analog Electronics

TECHNICAL SKILLS

Hardware and simulation: KiCad, LTspice, PCB layout, analog front ends, oscilloscopes, multimeters
Embedded: C/C++, STM32H7, ESP32, ATmega328P, ADC, DMA, timers, I²C, SPI, UART
Interfaces and data: Ethernet (W5500), WebSocket, BLE, Python, Flutter/Dart, CSV, Git

RESEARCH AND ENGINEERING EXPERIENCE

Undergraduate Research Intern | Iowa State University | Prof. Manojit Pramanik 2026–Present

- Developing an acquisition architecture for pulsed-LED photoacoustic tomography, evaluating STM32H7 timing, ADC triggering, DMA buffering, and a planned ESP32 data interface.
- Simulated photoacoustic analog conditioning in LTspice, comparing 1–20 MHz filter behavior under stated source and load assumptions; hardware response remains unmeasured.
- Drafted an eight-channel VCA8500 KiCad reference schematic and simulated TI channel-model gain versus control voltage; system integration, ERC cleanup, and hardware validation remain in progress.

Summer Research Intern | IIT Hyderabad | Prof. Rupesh Wandhere Jun–Jul 2025

- Built ESP32 display and Flutter monitoring prototypes using TFT/touch interfaces, BLE, WebSocket, and W5500 Ethernet connections.
- Connected four ESP32/W5500 nodes to independent Flutter WebSocket channels and a four-panel dashboard with per-device CSV recording for generated waveforms.
- Developed power-system interface and plotting workflows, including bounded chart histories and reconnect handling; public prototype telemetry uses simulated readings.

SELECTED PROJECTS

Photodiode TIA and Wireless Optical Sensor | OPA2320, ESP32, Python 2026

- Designed a selectable-gain photodiode analog front end with paired feedback components, filtering, and ADC protection; produced a two-layer PCB fabrication package.
- Simulated analog response and built ESP32 firmware and Python analysis tools; software checks passed, while PCB manufacture and hardware measurements remain pending.

MCU Data Logger PCB | ATmega328P, dual EEPROM, RTC 2026

- Designed and routed a two-layer KiCad Rev A board with dual 24LC1025 EEPROMs, a DS1337 RTC, UART/I²C access, and an ISP header.
- Completed schematic/ERC and PCB/DRC checks and generated Gerber/drill outputs; firmware, assembly, and physical bring-up remain pending.

ACTIVITIES

BITS Pilani Student Technical Team, App Development Vertical | IEEE Electrical Systems Vertical Competition finalist